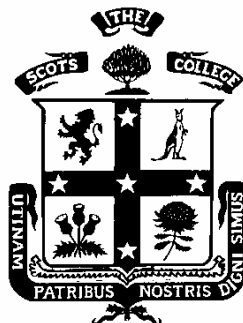


THE SCOTS COLLEGE



YEAR 12 EXTENSION 1 MATHEMATICS

PRE-TRIAL

APRIL 2006

WEIGHTING: 30%

TIME ALLOWED: 1½ HOURS (PLUS 5 MINUTES READING TIME)

INSTRUCTIONS TO STUDENTS:

- START EACH QUESTION IN A NEW BOOKLET.
- BOARD APPROVED CALCULATORS MAY BE USED.
- MARKS MAY BE DEDUCTED FOR CARELESS WORKING.
- ALL NECESSARY WORKING MUST BE SHOWN.

Question 1 (10 marks)

- a) Evaluate $\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$ 2
- b) Show that if p and q are the roots of the quadratic equation $2x^2 + 6x + 3 = m(2x + 1)$, then $p + q + 2pq = 0$ 2
- c) $\int xe^{x^2} dx$ 1
- d) If $\log_8 2 = \log_x 5$ find the value of x 2
- e) Solve the equation $\cos 2A = \cos A$ where $0 \leq A \leq 2\pi$ 3

Question 2 (12 marks)

- a) i) Find a point where the curve $y = x^3 - 4x^2 + 8x - 8$ crosses the x axis. 3
- ii) Show that this is the only point where the curve crosses the x axis.
- b)
- i. Sketch the graph of the function $y = \sin^{-1}\left(\frac{x}{2}\right)$ 1
- ii. State the domain and the range of the function 2
- iii. Find the exact equation of the tangent to the curve $y = \sin^{-1}\left(\frac{x}{2}\right)$ at the point where $x = 1$ 3
- c) The point $P(t + 3, t^2 - 5)$ lies on a parabola
- i. Find the Cartesian equation of the parabola 2
- ii. Find the vertex of the parabola 1

Question 3 (16 marks)

a)

i. Show that $\cos \sigma - \sqrt{3} \sin \sigma = 2 \cos \left(\sigma + \frac{\pi}{3} \right)$ **2**

ii. Hence solve the equation $\cos \sigma - \sqrt{3} \sin \sigma = 1$ for σ in the interval $0 \leq \sigma \leq 2\pi$ **2**

b) Use the substitution $u = \sqrt{x}$ to evaluate $\int_1^3 \frac{dx}{(1+x)\sqrt{x}}$ **3**
Give your answer in exact form.

c)

i. Differentiate $x \cos^{-1} x - \sqrt{1-x^2}$ with respect to x . **3**

ii. Hence, evaluate $\int_0^1 \cos^{-1} x \, dx$ **2**

d) $ABCD$ is a cyclic quadrilateral in which the opposite sides AB and DC are equal

i. Draw a diagram **1**

ii. Prove that the diagonals AC and BD are equal **3**

----- Question four and five continued on back page -----

Question 4 (11 marks)

- a) Find the sum of the first ten terms of the following Geometric series leaving your answer as a fraction.

$$4 - 2 + 1 - \frac{1}{2} + \dots$$

2

- b) Prove that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in arithmetic progression if $\frac{a}{c} = \frac{a-b}{b-c}$

3

- c) Find $\int \cos^2 x \, dx$

2

- d) The area enclosed between the curve $y = \sqrt{\frac{4}{1+x^2}}$ and the line $y = \sqrt{2}$ is rotated about the x axis. Find the volume of the solid of revolution generated. Give your answer in terms of π

4

Question 5 (13 marks)

- a) Use the method of mathematical induction to show that

4

$$1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

- b)

- i. Show that the equation of the tangent at the point $(2ap, ap^2)$ on the parabola $x^2 = 4ay$ is $y = px - ap^2$

2

- ii. This tangent cuts the x axis at Q . S is the focus. Prove that $\angle PQS$ is a right angle.

3

- iii. Show that the centre of the circle passing through P, Q and S is the midpoint of SP

2

- iv. Show that the locus of the centre of the circle through P, Q and S is $2ay = a^2 + x^2$

2

END OF PAPER

Question 1

a) $\cos^{-1}(-\frac{1}{2})$

$$= \pi - \cos^{-1}(\frac{1}{2}) \quad \checkmark$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3} \quad \checkmark$$

b) $2x^2 + 6x + 3 = 2m^2 + 2m$

$$2x^2 + 6x - 2mx + 3 - m = 0$$

$$2x^2 + x(6 - 2m) + 3 - m = 0$$

$$p+q = -\frac{B}{a}$$

$$pq = \frac{C}{a}$$

$$p+q = \frac{-6+2m}{2} \quad \checkmark$$

$$= \frac{3-m}{2}$$

$$= -3+m$$

$$\therefore p+q+2pq=0$$

$$(-3+m) + 2\left(\frac{3-m}{2}\right) = 0 \quad \checkmark$$

$$-3+m+3-m=0$$

$$0=0 \text{ As required}$$

c) $\int x e^{x^2} dx$

$$= \frac{1}{2} e^{x^2} + C$$

d) let $\log_8 2 = y$

$$2 = 8^y$$

$$2 = (2^3)^y$$

$$1 = 3y$$

$$y = \frac{1}{3} \quad \checkmark$$

$$\therefore \frac{1}{3} = \log_x 5$$

$$x^{\frac{1}{3}} = 5$$

$$x = 5^3$$

$$x = 125 \quad \checkmark$$

$$c) \cos 2A = \cos A$$

$$2\cos^2 A - 1 = \cos A$$

$$2\cos^2 A - \cos A - 1 = 0$$

$$\text{let } y = \cos A$$

$$2y^2 - y - 1 = 0$$

$$(2y+1)(y-1) = 0$$

$$y = -\frac{1}{2} \text{ or } y = 1$$

$$\therefore \cos A = -\frac{1}{2} \text{ or } \cos A = 1$$

$$A = 0^\circ, 120^\circ, 240^\circ, 360^\circ$$

$$0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi$$

QUESTION 2

$$a) y = x^3 - 4x^2 + 8x - 8$$

$$f(2) = 8 - 16 + 16 - 8 = 0$$

$\therefore (x-2)$ is a factor

$$\begin{array}{r} x^2 - 2x + 4 \\ x-2 \overline{) x^3 - 4x^2 + 8x - 8} \\ \underline{x^3 - 2x^2} \\ -2x^2 + 8x \\ \underline{-2x^2 + 4x} \\ 4x - 8 \\ \underline{4x - 8} \\ 0 \end{array}$$

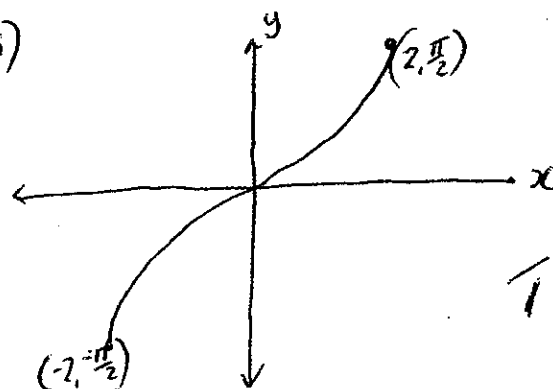
$$x^3 - 4x^2 + 8x - 8 = (x-2)(x^2 - 2x + 4)$$

$$\Delta = 4 - 4(1)(4)$$

$$\Delta = -12 \text{ No solution}$$

$$\therefore x = 2$$

b) i)



$$ii) \text{ Domain } -2 \leq x \leq 2$$

$$\text{Range } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$iii) y = \sin^{-1}\left(\frac{x}{2}\right)$$

$$y' = \frac{1}{\sqrt{1 - \frac{x^2}{4}}} \cdot \frac{1}{2}$$

$$= \frac{1}{\sqrt{\frac{4-x^2}{4}}} \cdot \frac{1}{2}$$

$$= \frac{2}{\sqrt{4-x^2}} \cdot \frac{1}{2}$$

$$y' = \frac{1}{\sqrt{4-x^2}}$$

$$\text{when } x=1 \quad m = \frac{1}{\sqrt{4-1}}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{3}$$

$$\text{when } x=1 \quad y = \sin^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{\pi}{6}$$

$$\frac{1}{2}$$

Question 2 (cont)

$$m = \frac{\sqrt{3}}{3} \quad \text{pt } (1, \frac{\pi}{6})$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{6} = \frac{\sqrt{3}}{3}(x - 1)$$

$$6y - \pi = 2\sqrt{3}(x - 1)$$

$$6y - \pi = \cancel{2\sqrt{3}x} - 2\sqrt{3}$$

$$2\sqrt{3}x - 6y + \pi - 2\sqrt{3} = 0$$

c) i) $x = t + 3$

$$x - 3 = t$$

$$y = (x - 3)^2 - 5$$

ii) $(x - 3)^2 = (y + 5)$

$$V = (3, -5)$$

Question 3

$$\begin{aligned} \text{a) i) } 2 \cos \left(\theta + \frac{\pi}{3} \right) &= 2 \left(\cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right) \quad \checkmark \\ &= 2 \left(\cos \theta \cdot \frac{1}{2} - \sin \theta \cdot \frac{\sqrt{3}}{2} \right) \\ &= \cos \theta - \sqrt{3} \sin \theta \quad \text{As required} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii) } 2 \cos \left(\theta + \frac{\pi}{3} \right) &= 1 \quad \checkmark \\ \cos \left(\theta + \frac{\pi}{3} \right) &= \frac{1}{2} \\ \theta + \frac{\pi}{3} &= \frac{\pi}{3} \text{ or } \frac{5\pi}{3} \text{ or } \frac{7\pi}{3} \\ \theta &= 0 \text{ or } \frac{4\pi}{3} \text{ or } 2\pi \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } u &= x^{\frac{1}{2}} \\ \frac{du}{dx} &= \frac{1}{2} x^{-\frac{1}{2}} \\ du &= \frac{1}{2\sqrt{x}} dx \rightarrow 2x^{\frac{1}{2}} du = dx \quad \checkmark \end{aligned}$$

$$\begin{aligned} &\int \frac{2x^{\frac{1}{2}} du}{x^{\frac{1}{2}} \cdot 1+u^2} \\ &= \int_1^3 \frac{2}{1+u^2} du \\ &= 2 \left[\tan^{-1} u \right]_1^{\sqrt{3}} \quad \checkmark \\ &= 2 \left(\tan^{-1} \sqrt{3} \right) - \left(\tan^{-1} 1 \right) \\ &= 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= 2 \times \frac{\pi}{12} \\ &= \frac{\pi}{6} \quad \checkmark \end{aligned}$$

c) $x \cos^{-1} x - \sqrt{1-x^2}$

Let $t = x \cos^{-1} x$

$$\frac{dt}{dx} = x \cdot \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x$$

$$= \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x$$

✓

$$\therefore y' = \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x - \frac{1}{2}(1-x^2)^{-1/2} \cdot -2x$$

✓

$$y' = \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x + \frac{x}{\sqrt{1-x^2}}$$

✓

$$y' = \cos^{-1} x$$

ii) $\int_0^1 \cos^{-1} x \, dx$

$$= \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_0^1$$

✓

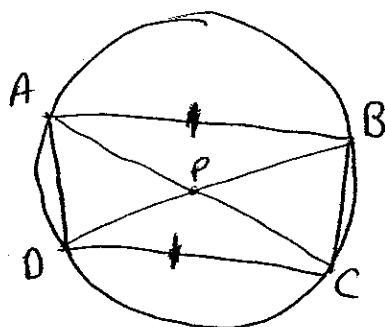
$$= \left[1 \cos^{-1} 1 - \sqrt{1-1^2} \right] - \left[0 \cos^{-1} 0 - \sqrt{1-0^2} \right]$$

$$= \left[0 \right] - \left[0-1 \right]$$

✓

$$= 1$$

d)



✓

In Δ 's ABP and DCP

$\angle ABP = \angle DCP$ (l's in same segment)

$\angle CDP = \angle BAP$ (l's in same segment) ✓

$AB = DC$ (data)

$\therefore \Delta ABP \equiv \Delta DCP$ (ASA)

$AP = PD$ (Corresponding sides in Congruent Δ 's)

$BP = PC$ (Corresponding sides in Congruent Δ 's)

$$AP + PC = BP + PD$$

✓

Question 4

a) $a=4$ $r=-\frac{1}{2}$ $n=10$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$= \frac{4(1-(-\frac{1}{2})^{10})}{1+\frac{1}{2}}$$

$$= \frac{4(1-\frac{1}{1024})}{\frac{3}{2}}$$

$$= \frac{8}{3} \times \frac{1023}{1024}$$

$$= 2\frac{85}{128}$$

b) $\frac{a}{c} = \frac{a-b}{b-c}$

$$ab-ac = ac-bc$$

$$\div abc) \quad \frac{ab}{abc} - \frac{ac}{abc} = \frac{ac}{abc} - \frac{bc}{abc}$$

$$\frac{1}{c} - \frac{1}{b} = \frac{1}{b} - \frac{1}{a}$$

which is proof of AP

c) $\cos 2x = 2\cos^2 x - 1$

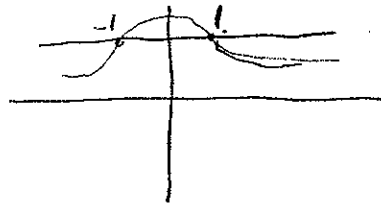
$$\cos 2x + 1 = 2\cos^2 x$$

$$\frac{1}{2} [\cos 2x + 1] = \cos^2 x$$

$$\therefore \frac{1}{2} \int \cos 2x + 1 \, dx$$

$$= \frac{1}{2} \left[\frac{\sin 2x}{2} + x \right] + C$$

d)



$$\sqrt{2} = \sqrt{\frac{4}{1+x^2}}$$

$$2x^2 + 2 = 4$$

$$2x^2 - 2 = 0$$

$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$x = 1, \text{ or } -1$$

If $y = \sqrt{\frac{4}{1+x^2}}$

$$y^2 = \frac{4}{1+x^2}$$

$$2\pi \int_0^1 \frac{4}{1+x^2} \, dx$$

$$8\pi \left[\tan^{-1} x \right]_0^1$$

$$= 8\pi \left[\frac{\pi}{4} - 0 \right]$$

$$= 2\pi^2 \text{ units}^3$$

$$y = \sqrt{2}$$

$$y^2 = 2$$

$$2\pi \int_0^1 2 \, dx$$

$$2\pi \left[2x \right]_0^1$$

$$2\pi \left[2 - 0 \right]$$

$$= 4\pi$$

$\therefore \text{Volume} = 2\pi^2 - 4\pi$

$$= 2\pi \left[\pi - 2 \right] \text{ units}^3$$

Question 5

a)

When $n=1$

$$1^2 = \frac{1}{6}(1+1)(2+1)$$

$$= \frac{1}{6} \times 2 \times 3$$

$$1 = 1$$

$\frac{1}{2}$

Assume true for $n=k$ and show true for $n=k+1$

$$S(k) = 1^2 + 2^2 + 3^2 + \dots + k^2$$

$$S(k+1) = 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$$

Assume that $S(k) = \frac{1}{6}k(k+1)(2k+1)$ and show that

1

$$S(k+1) = \frac{1}{6}(k+1)(k+2)(2k+3)$$

$$\text{Now } S(k+1) = S(k) + (k+1)^2$$

$$= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$= (k+1) \left[\frac{1}{6}k(2k+1) + (k+1) \right]$$

$$= \frac{1}{6}(k+1)(k(2k+1) + 6(k+1))$$

$$= \frac{1}{6}(k+1)(2k^2 + k + 6k + 6)$$

$$= \frac{1}{6}(k+1)(2k^2 + 7k + 6)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3)$$

2

if the result is true for $n=k$, then it is true for $n=k+1$ $\frac{1}{2}$

$$b) i) x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$y' = \frac{x}{2a}$$

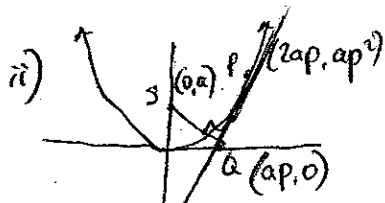
$$\text{when } x = 2ap \quad m = \frac{2ap}{2a}$$

$$m = p$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$



Co-ords of Q

$$y = px - ap^2$$

$$0 = px - ap^2$$

$$xp = ap^2$$

$$x = ap$$

$$\therefore Q(ap, 0)$$

$$\text{Gradient of } PQ = p$$

$$\text{Gradient of } SQ = \frac{a-0}{0-ap}$$

$$= \frac{a}{-ap}$$

$$= -\frac{1}{p}$$

$$\text{For Right Angle } m_1 \times m_2 = -1$$

$$p \times -\frac{1}{p} = -1$$

$$-1 = -1$$

So PQ is Perp to SQ

iii) Since $\hat{PQS}$ is a Right Angle $\therefore$ A Semi-circle can be drawn through P, Q and S, Since the angle in a semi-circle is a right angle $\therefore$ PS is a Diameter to the circle and so the midpoint of PS must be the centre of the circle.

iv) Co-ords of Midpoint of PS.

$$x = \frac{0+2ap}{2}$$

$$x = ap$$

$$y = \frac{a+ap^2}{2}$$

$$= \frac{a(1+p^2)}{2}$$

$$\text{From } x = ap$$

$$\frac{x}{a} = p$$

$$\therefore y = \left[a \left(1 + \frac{x}{a} \right)^2 \right] \div 2$$

$$= a \left(1 + \frac{x^2}{a^2} \right) \div 2$$

$$= \left(a + \frac{x^2}{a} \right) \div 2$$

$$= \frac{a^2 + x^2}{2a}$$

$$y = \frac{a^2 + x^2}{2a}$$

$$2ay = a^2 + x^2 \quad \text{As required}$$